

Ritabrata Chakraborty

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EDUCATION

University of Pennsylvania <i>M.S.E. in Robotics, GPA: -/4</i>	Pennsylvania, USA <i>Aug '26 – May '28</i>
Birla Institute of Technology and Science, Pilani (BITS Pilani) <i>B.E. in Mechanical Engineering — Minor in Data Science, CGPA: 8.57/10</i> • Institute Merit-cum-Need Scholarship — Consecutive Recipient (Top 8% of Students)	Rajasthan, India <i>Oct '22 – May '26</i>

LEADERSHIP & VOLUNTEERING

Deputy Director of Academic Programming <i>Graduate and Professional Student Assembly (GAPSA), University of Pennsylvania</i>	<i>Aug '26 – Present</i>
Vice-President <i>South Asian Association (Rangoli), University of Pennsylvania</i>	<i>Aug '26 – Present</i>
President & Secretary <i>Mechanical Engineering Association (MEA), BITS Pilani</i> • Coordinated 10+ events and career sessions for 300+ students, facilitating technical exposure and alumni interaction. • Managed production of 500+ merchandise items and led outreach, boosting student participation by 20%.	<i>Jun '24 – May '26</i>
President & Tech Fest Coordinator <i>Indian Society of Heating, Refrigerating, and Air Conditioning Engineers (ISHRAE), BITS Pilani</i> • Led a team of 25+ members to organize 4+ technical workshops with HVAC industry experts. • Hosted 3 competitions and networking events, engaging 200+ students in HVAC innovation and awareness.	<i>Oct '24 – Jul '25</i>
Project Manager <i>Tinkerer's Lab (TL), BITS Pilani</i> • Supervised 5 interdisciplinary robotics teams (30+ members) on Micromouse and Hexapod projects. • Oversaw lab resources, conducted weekly reviews, and mentored 50+ students in hands-on technical skills.	<i>May '24 – Jul '25</i>
Treasurer <i>Bengali Cultural Association (Moruchhaya), BITS Pilani</i> • Allocated funds and maintained financial records for merchandise sales, cultural programs, and annual events.	<i>May '24 – Jul '25</i>
Volunteer <i>Junoon, National Service Scheme (NSS), BITS Pilani</i> • Organized sports and recreational competitions for specially abled children from 25+ colleges.	<i>Sep '23 – Sep '23</i>

TEACHING

Teaching Assistant <i>CIS 4190/5190: Applied Machine Learning, University of Pennsylvania</i>	<i>Aug '26 – Dec '26</i>
<i>CS F320: Foundations of Data Science, BITS Pilani</i>	<i>Jan '26 – May '26</i>
<i>ME F218: Advanced Mechanics of Solids, BITS Pilani</i>	<i>Jan '25 – May '25</i>
<i>ME F216: Materials Science and Engineering, BITS Pilani</i> • Assisted 100+ students in labs and tutorials, clarifying concepts and linking theory to practical applications. • Analyzed assignments and facilitated faculty in delivering high-impact teaching sessions.	<i>Sep '24 – Dec '24</i>

PUBLICATIONS & PEER REVIEWS

Peer Reviews • Selected as a peer reviewer for IEEE Gujarat Section Conference , 2026.

INDUSTRY EXPERIENCE

ML Intern Uber (AI Solutions) <i>Supervisor: Mr. Siddarth Malreddy, Tech Lead Manager & Mr. Ishan Nigam, Senior ML Engineer, Uber</i> • Streamlined object tracking in uLabel for RGB/IR videos, flagging object annotation anomalies via XGBoost (mAP50: 75%). • Screened object-association errors with a TrackTrack -based cross-frame consistency check (F1: 88%).	<i>Jul '25 – Dec '25</i>
Subject Matter Expert Knowledge Cultivator <i>Self-supervised</i> • Resolved numerical problems for student clients using MATLAB, CAD, and Python. • Authored academic content across diverse research topics.	<i>Nov '23 – Feb '24</i>

AWARDS & ACHIEVEMENTS

Finalist — Meta PyTorch OpenEnv Hackathon (India) 🔗	Apr '26
3 rd Place — MathWorks Global Drone Student Challenge 2025	Mar '25
Finalist — ISRO Immersion Challenge (AI for Space & Geospatial Innovation) 🔗	Jul '24
Top 15 Overall, Top 5 in College — American Express Campus Challenge 2024 🔗	Jul '24
5 th Place & Best Maintenance Award — European Rover Challenge 2023 (Remote Finals)	Sep '23

PROFESSIONAL ACTIVITIES

Graduate Student Member — IEEE	2026
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COMPETITIVE EXAMINATIONS

GRE: 317 (Quantitative: 168, Verbal: 149, AWA: 3)	2025
TOEFL iBT: 108 (Reading: 27, Listening: 30, Speaking: 23, Writing: 28)	2025
JEE Advanced: All India Rank 14372	2022

SKILLS

Leadership & Communication: Event Planning, Team Management, Budget Allocation, Mentorship, Cross-Functional Coordination, Public Speaking, Stakeholder Outreach, Technical Writing
Programming Languages: Python, C++, C, C#, Java, Bash (Shell)
Productivity Software: Microsoft Office (Excel, PowerPoint, Word), Google Workspace